Program: **Diploma in Automation and Robotics**	
Course Code: **6332D**	Course Title: **Digital Signal Processing**
Semester : **6**	Credits : **4**
Course Category : **Open Elective**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To understand different types of signals and signal operations.
- To understand different classification of systems
- Tounderstand Fourier and Z Transform
- Tounderstand Fast Fourier transform
- TounderstandDigital signal Processors

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge in mathematics		Mathematics1and Mathematics II	
Basics of electrical networks		Fundamentals of Electrical circuits and machines ,Electrical circuits and Machines	2,3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understanddifferenttypesofsignalsandsignaloperations and Classification of Systems	14	Understanding
CO2	Understand Fourier and Z Transform	15	Understanding
CO3	Understand Fast Fourier transform	14	Understanding

CO4	UnderstandDigital signal Processors	15	Understanding and Applying
	Series Test	2	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3	2					
CO3			3				
CO4	3						

3-Strongly mapped 2-Moderately mapped 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand different types of signals and signal operations and Classification of Systems**		
M1.01	Describe Discrete time signals. Define unit impulse, unit step, ramp and exponential signals Sinusoidal signals, periodic and a periodic signals ,Even and odd signal, causal and non causal signal	4	Understanding
M1,02	Explain Shifting, time reversal and times caling operation Scalar multiplication and signal multiplication and Addition operator	3	Understanding
M1.03	To explain discrete time system, linear and non linear, causal and non causal system To describe Time variant and time invariant system	3	Understanding
M1.04	To define LTI system , block diagram of a DSP system To list the advantage of DSP system	4	Understanding
Contents: Classification of signals- discrete time signals- unit step –unit ramp-unit impulse, exponential sequence, sinusoidal signal - periodic and non periodic signals, even and odd signals - causal and non- causal signal, operation of signal –shifting, time reversal , time scaling, scalar multiplication, signal multiplier, addition, discrete time system – classification – linear and nonlinear system – causal and non-causal system – time variant and time invariant system, LTI system - block diagram of a DSP system – advantages of a DSP system			

CO2	**Understand Fourier and Z Transform**		
M2.01	DescribediscreteFourierseriesand discreteFouriertransform	2	Understanding
M2.02	Find DFT of unit impulse and other dc signal, sinusoidal signal and properties of DFT, State Linearity and periodicity, circular shift and symmetry	4	Understanding
M2.03	Explaincircularconvolution	2	Applying
M2.04	Find Z transform of unit step, unit impulse and sinusoidal signal	2	Understanding
M2.05	Discuss properties of Z transform; linearity, left shift, right shift	2	Understanding
M2.06	Discuss inverse Z transform and list the types inverse Z transform	3	Applying
	Series Test – I	1	
Contents: Fourier series- fundamentals - Discrete Fourier transform- DFT of unit impulse, unit step signal, sinusoidal signal, properties of DFT- linearity- periodicity- circular shift – symmetry property- circular convolution of time domain signal- circular convolution of frequency domain ,Z- transform- Z-transform of unit step- unit impulse – sinusoidal, property of z- transform- linearity – left shift of a signal- right shift of a signal- convolution – multiplication by anu(n) – initial value theorem – final value theorem , inverse z-transform			
CO3	**Understand Fast Fourier transform**		
M3.01	Understand Fast Fourier Transform and Decimation in time	1	Understanding
M3.02	To understand 4 point and 8 point FFT using radix 2 DIT block diagram	4	Understanding
M3.03	Draw 8 point FFT using radix 2 DIT butterfly diagram	2	Applying
M3.04	Understand decimation in frequency 4 point and 8 point FFT using radix 2DIF block diagram	4	Understanding
M3.05	Draw 8 point FFT using radix 2DIF butterfly diagram.	3	Applying
CONTENTS			
Contents: Decimation in Time(DIT) – 4 point and 8 point FFT using radix 2 DIT FFT- flow graph for 8 point DFT, Decimation in Frequency(DIF)- 4 point and 8 point FFT using radix 2 DIF FFT, comparison of DIT and DIF.			
CO4	**Understand Digital signal Processors**		

M4.01	Understand FIR filters and explain about FIR filter coefficients	3	Understanding
M4.02	Describe IIR filters and IIR filter coefficients	3	Understanding
M4.03	Overview of Digital signal processors and selection of such processors	3	Understanding
M4.04	Understand architectural features, execution speed, type of arithmetic, word length	3	Understanding
M4.05	Describe the Texas DS processor TMX320c50DSP and its addressing modes	3	Understanding
	Series Test – II	1	
Contents: Filters – types - FIR filter – coefficients - FIR windows, IIR filters – coefficients, Overview of a digital signal processor – selecting a digital signal processor- architecture of Texas Instruments- TMX320c50 DSP – CPU – Central ALU – Parallel Logic Unit – Auxiliary Register -Arithmetic unit- index Register- Aux. Reg Compare Reg. - Block move address register – status register- program controller- program counter – on chip memory – on chip peripherals, addressing modes, Applications of DSP			

Text / Reference

T/R	Book Title/Author
T1	DigitalSignalProcessing–Salivahanan–TMH-2ndEdition
T2	DigitalSignalProcessing-PRameshBabu,Scitech-4thEdition
R1	DigitalSignalProcessing–NagoorKani–TMH
R2	DigitalSignalProcessing–AlanVOppenheim,RonaldWSchafer(Pearson)
R3	A Textbook of Digital Signal Processing–RS Kaler, M Kulkarni, Umesh Gupta (IK International publishing company, NewDelhi)

Online Resources

Sl.No	WebsiteLink
1	https://onlinecourses.nptel.ac.in/noc23_ee33/preview